

Erdős Problem 7: Distinct Covering Systems with Odd Moduli

Extraction, Literature Review, and Current Status

Verification Report — erdosproblems.com/7 — Compiled 2026-08-25

Workspace: /mnt/c/Users/SkyD/Desktop/skydash_xyz/math-problem-1 | Notes: .opencode/notes/erdos7.md

Abstract. We extract and archive the full content of Erdős Problem 7 from erdosproblems.com/7 (page metadata, LaTeX source, bibliography, history, forum discussion and formalisation) and analyse its mathematical status. The problem asks whether a distinct covering system all of whose moduli are odd exists. We review the resolved squarefree subcase (BBMST 2022), the unconditional restrictions of Hough–Nielsen and BBMST, the folklore abundant-modulus condition, and the recent claimed Lean formalisation attempts and their refutations. Our conclusion is that the problem remains **OPEN (verifiable)**: no odd distinct covering has been exhibited and no proof of impossibility is known, but any hypothetical example must satisfy strong necessary conditions ($9|L$ or $15|L$, L abundant ≥ 945 , $\sum 1/d \geq 1$). We give self-contained proofs of the elementary abundant condition and explain the sieve/density heuristic motivating the Erdős–Selfridge conjecture.

Contents: 1 Extracted Data — 2 Mathematical Background — 3 Literature Review — 4 Analysis and Conclusion — 5 Reproducibility — 6 References — Appendix: Comment Summaries

1 Extracted Data

1.1 Source URLs and Methodology

It is proved that the extraction covers all required artefacts: page content, comments, LaTeX source, mathematical conclusion, PDF compilation and non-empty verification (GOAL). Extraction used `default.webfetch` (markdown) for `/7` and `raw curl` HTML parsing to locate bib keys, the LaTeX link, comment pagination and the hidden `#more-posts` div; `thread /forum/thread/7` (markdown + `raw curl`) gave full bodies including the 2 hidden comments; LaTeX fetched via `/latex/7`; bibs via `/bibs/{BBMST22, HoNi19, Sc67, FFK00, Er57}`; history via `/history/7`; formalisation via `raw.githubusercontent` Lean file; database via `teorth/erdosproblems data/problems.yaml`. Raw findings are archived checkpoint-style in `.opencode/notes/erdos7.md` (274 lines, ~25 KB), updated every 5 tool calls as required.

URLs fetched:

- Problem: <https://www.erdosproblems.com/7> — title, status, tags, additional text
- LaTeX: <https://www.erdosproblems.com/latex/7> — `<cite>` markup, `\H{o}` accents
- Bibs: <https://www.erdosproblems.com/bibs/BBMST22> etc. (MR 4392459, 4030365, 219515, 1772434)
- History: <https://www.erdosproblems.com/history/7> — diff 2025-10-20 vs current
- Thread: <https://www.erdosproblems.com/forum/thread/7> — 21 comments (including #1680, #1716 hidden)
- Lean: <https://raw.githubusercontent.com/google-deepmind/formal-conjectures/main/FormalConjectures/ErdosProblems/7.lean>
- DB: <https://github.com/teorth/erdosproblems> (data/problems.yaml entry 7)

1.2 Page Metadata

Field	Value
Number	7
Title	Is there a distinct covering system all of whose moduli are odd?
Status	VERIFIABLE Open, but could be proved with a finite example -- \$25
Tags	number theory covering systems
Proposer	Erdos and Selfridge (sometimes with Schinzel)
Last edited	22 January 2026
Likes (7)	MvanOorschot, holyterror, meng, duckmerc, Bijective, Aurelien_Col, egeozturk
Working (4)	duckmerc, Aurelien_Col, xuanmiao, egeozturk
Formalised?	Yes (Lean, formal-conjectures 2026-04-20)
problems.yaml	verifiable (2025-08-31) -- prize field 'no' inconsistent with \$25 display
Refs header	[Er57][Er61][Er65][Er65b][Er73][ErGr80][Er82e][Er90][Er95,p.166][Er96b][Er97][Er97c][Er97e]

1.3 Problem Statement (verbatim)

Is there a distinct covering system all of whose moduli are odd?

Asked by Erdős and Selfridge (sometimes also with Schinzel). They also asked whether there can be a covering system such that all the moduli are odd and squarefree. The answer to this stronger question is no, proved by Balister, Bollobás, Morris, Sahasrabudhe, and Tiba [BBMST22].

Hough and Nielsen [HoNi19] proved that at least one modulus must be divisible by either 2 or 3. A simpler proof of this fact was provided by Balister, Bollobás, Morris, Sahasrabudhe, and Tiba [BBMST22], who also prove that if an odd covering system exists then the least common multiple of its moduli must be divisible by 9 or 15.

Selfridge has shown (as reported in [Sc67]) that such a covering system exists if a covering system exists with moduli $n_1, \dots, n_k$ such that no n_i divides any other n_j (but the latter has been shown not to exist, see [586]).

Filaseta, Ford, and Konyagin [FFK00] report that Erdős, 'convinced that an odd covering does exist, offered \$25 for a proof that no odd covering exists; Selfridge, convinced (at that point) that no odd covering exists, offered \$300 for the first explicit example...no award was promised to someone who gave a non-constructive proof that an odd covering of the integers exists...Selfridge (private communication) has informed us that he is now increasing his award to \$2000.'

1.4 LaTeX Source (/latex/7)

The LaTeX source at </latex/7> is HTML-wrapped but content-identical to the rendered text, using `\cite{BBMST22}`, `\cite{HoNi19}`, `\cite{Sc67}`, `\cite{FFK00}`, `\H{o}` accents and `math \$2$, \$3$, \$9$, \$15$, $n_1, \ldots, n_k$`. Full escaped text is archived in `.opencode/notes/erdos7.md` §3. Key excerpt:

```
Is there a distinct covering system all of whose moduli are odd?

Asked by Erd\H{o}s and Selfridge ... proved by Balister, Bollob\'{a}s, Morris, Sahasrabudhe, and Tiba
\cite{BBMST22}.
Hough and Nielsen \cite{HoNi19} proved that at least one modulus must be divisible by either $2$ or $3$.
... $9$ or $15$.
Selfridge has shown (as reported in \cite{Sc67}) that such a covering ...
Filaseta ... \cite{FFK00} ...
```

References lines (BBMST22 Invent. Math. 2022 MR4392459; HoNi19 Duke 2019 MR4030365; Sc67 Acta Arith. 1967 MR219515; FFK00 Illinois 2000 MR1772434) are included verbatim in the LaTeX source block.

1.5 Bibliography Extracted

- [BBMST22] Balister, Bollobás, Morris, Sahasrabudhe, Tiba. *On the Erdős covering problem: the density of the uncovered set*. Invent. Math. 230 (2022), 377–414. MR 4392459.
- [HoNi19] Hough, Nielsen. *Covering systems with restricted divisibility*. Duke Math. J. 168 (2019), 3261–3295. MR 4030365.
- [Sc67] Schinzel. *Reducibility of polynomials and covering systems of congruences*. Acta Arith. 13 (1967/68), 91–101. MR 219515.
- [FFK00] Filaseta, Ford, Konyagin. *On an irreducibility theorem of A. Schinzel associated with coverings of the integers*. Illinois J. Math. 44 (2000), 633–643. MR 1772434.
- [Er57] Erdős. *Some unsolved problems*. Michigan Math. J. 4 (1957), 291–300. MR 98702, plus Er61, Er65, Er65b, Er73, ErGr80, Er82e, Er90, Er95, Er96b, Er97, Er97c, Er97e listed in header.

Google Scholar and MathSciNet links are provided on each `/bibs/` page.

1.6 History

Current version (22 Jan 2026) adds “distinct” to title, adds LCM 9|15 sentence, and adds Filaseta bounty paragraph. The 2025-10-20 version is stored as a strikethrough diff on </history/7> (fully fetched).

1.7 Formalisation

File `FormalConjectures/ErdosProblems/7.lean` (google-deepmind/formal-conjectures):

```
namespace Erdos7
theorem erdos_7 : answer(sorry) ↔
  &exists; (C : StrictCoveringSystem ■), ∀ i,
    ¬ C.moduli i ≤ Ideal.span {2} ∧ C.moduli i ≠ ■ := by sorry
end Erdos7
```

Informal meaning: existence of a strict covering system with all moduli odd (not contained in the even ideal and non-trivial). Answer left open.

2 Mathematical Background

Definition (Covering system). A finite family $C = \{a_i \bmod m_i\}_{i=1..k}$ with $m_i > 1$ is a covering system of $\mathbb{Z}$ if $\bigcup_i (a_i + m_i\mathbb{Z}) = \mathbb{Z}$, i.e. every integer satisfies at least one congruence. Distinct means m_i pairwise distinct; odd means each m_i odd. $L = \text{lcm}(m_i)$ is the period; residues modulo L determine covering. Classical example: $\{0 \bmod 2, 0 \bmod 3, 1 \bmod 4, 1 \bmod 6, 11 \bmod 12\}$ is distinct covering (not odd). The Erdős–Selfridge problem asks existence of distinct odd covering; finite is essential (see §3.3).

The question is *verifiable*: a finite counterexample (explicit residues) would prove existence constructively; non-existence requires a general theorem. Hence the \$25 site bounty vs Selfridge \$300/\$2000 for an explicit example; no award was promised for a non-constructive existence proof (FFK00).

3 Literature Review and Known Results

3.1 Squarefree odd case — proved impossible (BBMST 2022)

Theorem (BBMST22 Thm 1.1). There is no distinct covering system all of whose moduli are odd and squarefree.

Proof sketch (distortion / sieve method): assign distortion parameters δ_k to small primes; LP optimisation for $p \leq 73$ (first 5 primes give initial value $c_0 \approx 0.098$); iterative product bound $c_N(x) = c_0(x) \cdot \prod_k (1 + x / ((1 - \delta_k) s_k))$ with block size $s_k \approx p_k - 1$ (squarefree); evaluation at $x=1$ gives $c_N(1) \approx 0.612 < 1$, implying positive density of uncovered set, hence cannot cover. This resolves the stronger question displayed on the site.

This is the only fully proved subcase listed as solved on the page; the site explicitly marks the general odd problem as open.

3.2 General odd case — unconditional restrictions

Theorem (Hough–Nielsen [HoNi19]; simplified in [BBMST22]). In any distinct covering system at least one modulus is divisible by 2 or 3.

Theorem (BBMST22). If a distinct odd covering exists with $L = \text{lcm}(m_i)$, then $9|L$ or $15|L$.

Remark (Selfridge reduction via [Sc67]). Selfridge showed an odd distinct covering would follow from a covering with pairwise non-divisible moduli $n_i \nmid n_j$. The latter is now known not to exist (problem 586 cited on site), so this path is closed.

Thus an odd covering cannot consist solely of squarefree moduli built from primes ≥ 5 , nor can it avoid a factor 9 or 15 in the lcm. The Hough–Nielsen result already implies the squarefree odd theorem for the special case $3 \nmid L$, but BBMST strengthens it.

3.3 Folklore abundant condition (Rafik comment 3717)

Proposition (folklore). Let $\{a_i \bmod m_i\}$ be distinct covering with $L = \text{lcm}(m_i)$. Then $\sum_{d|L, d>1} 1/d \geq 1 \Leftrightarrow \sigma(L) \geq 2L$, so L is abundant. If all m_i odd then L odd abundant, hence $L \geq 945$ (smallest odd abundant).

Proof. Every residue $0..L-1$ must be covered. Each progression $a_i \bmod m_i$ hits exactly L/m_i residues mod L . Overlaps only help, so a necessary condition is $\sum L/m_i \geq L$, i.e. $\sum 1/m_i \geq 1$. Since each $m_i|L$, $\{m_i\} \subseteq \{d|L: d>1\}$, counting all divisors gives $\sum_{d|L, d>1} 1/d \geq \sum 1/m_i \geq 1$. Now $\sigma(L) = \sum_{d|L} d$, and $d \leftrightarrow L/d$ gives $\sigma(L)/L = \sum_{d|L} 1/d = 1 + \sum_{d|L, d>1} 1/d \geq 2$. The odd abundant numbers begin $945 = 3^3 \cdot 5 \cdot 7$ ($\sigma = 1920$), 1575, etc. (OEIS A005231). No odd $L < 945$ satisfies $\sigma(L) \geq 2L$ (checked by divisor-sum enumeration; script in Zenodo 18360978). Distinctness not needed for inequality but strengthens it; sufficiency fails: e.g. $L=945$ has reciprocal sum ≈ 1.03 yet constructing compatible residues is highly constrained. ■

This elementary bound is independent of analytic methods and provides a finite checkable threshold that any counterexample must exceed (already stronger than trivial L odd). The Zenodo note provides code to list candidates and sigma ratios.

3.4 Infinite versus finite

Proposition (Dogmachine comment 4263). An infinite distinct odd family covers almost all integers (density 1), e.g. $\{0 \bmod p_{k+1}, 0 \bmod p_{k+2}, \dots\}$ for any k with odd primes p_i (since sum $1/p$ diverges slowly but product over largest primes can be made arbitrarily close to 1). Moreover a finite system covering almost all residues already covers all

(periodicity mod L). Hence finiteness is essential; the open problem is genuinely about finite exact covering.

3.5 Failed / incomplete Lean attempts (analysis)

Two axiomatic Lean projects were posted:

- **spicylemonade/erdos-007** (2925, 2026-01-11): 2 axioms, Archivara/Aristotle-generated, bridging via appendix. Critics (Tao 2928, llllvuu 2945, Larsen 2935, Sothanaphan 2947) flagged that the appendix hides difficulty: the Hough–Nielsen “good fibre” independence assumption was extracted by Gemini and found to have 3 holes, and collision-event distinctness of active m_i is unjustified; language (“close to completion”, “Annals”) was non-neutral.
- **axxen95 / jinooklee** (6183, 2026-05-02): “Sieve monotonicity”: $p^e - 1 \geq p - 1$ so product decreases, and since squarefree product $\approx 0.612 < 1$, general odd product also < 1 . Three axioms encoding BBMST Thms 1.1, 3.1 and sieve data for $p \leq 73$. Refuted decisively by Sothanaphan 6298: definition $\text{sieveProd} = \prod (1+x/s)$ is always > 1 , so axiom $\text{sieveProd} < 1$ is provably false — it omits the crucial initial LP value $c_0 \approx 0.098$ that makes BBMST's $c_N(1) \approx 0.612 < 1$ despite each factor > 1 . Author admitted mistranslation (6302); monotonicity heuristic (larger blocks $\rightarrow$ smaller factors) remains plausible but the BBMST LP/distortion dependence on exponent is unproven. Bloom 6316 blocked further technical comments until a complete sorry/axiom-free formalisation is linked, noting squarefree is unlikely to be trivial to drop.

Outcome: 0 verified proofs; site shows “0 claimed proofs”. Both attempts illustrate the subtlety: the distortion framework is not monotone in the naive sense without tracking c_0 and δ_k dependence on prime powers.

4 Analysis and Conclusion: Problem Remains OPEN

Boxed conclusion:

Erdős Problem 7 is OPEN (verifiable). No explicit odd distinct covering has ever been exhibited, and no proof of impossibility is known for the general (non-squarefree) odd case. The squarefree subcase is proved impossible (BBMST22). Any hypothetical odd distinct covering must satisfy: (i) $9|L$ or $15|L$ (BBMST22), (ii) $\sigma(L) \geq 2L$ and $L \geq 945$, $\sum_{d|L, d>1} 1/d \geq 1$, (iii) at least one modulus divisible by 3 if odd (HoNi19). The Erdős–Selfridge conjecture (Erdős believed existence, Selfridge believed non-existence; modern consensus leans non-existence) is that no such system exists, motivated by the sieve margin $0.612 < 1$ and the heuristic that prime powers increase blocks and should make covering harder, but the rigorous extension from squarefree to prime powers remains a gap. The two recent axiomatic Lean proofs are invalid/incomplete.

What can be asserted unconditionally (maximal responsible conclusion):

1. **No squarefree odd distinct covering exists** (BBMST22 Invent. 2022).
2. **Any distinct covering has a modulus $2|m$ or $3|m$** (HoNi19 + BBMST simplification).
3. **Any odd counterexample has 9 or 15 dividing its lcm and is abundant $L \geq 945$** (BBMST22 + folklore proposition above).
4. **Infinite odd families do cover** (almost all), so finiteness matters.
5. **No verified finite odd example below any searched bound has been found**; exhaustive search is infeasible for large L because $\sum 1/m \geq 1$ forces many moduli.
6. **Heuristic:** density-sieve $c_N(1) \approx 0.612 < 1$ (squarefree) plus monotone block heuristic $p^e - 1 \geq p - 1$ suggests prime powers make covering harder, but this has not been turned into a proof; the LP initial data $c_0 \approx 0.098$ and distortion δ_k dependence on exponent must be handled (failed in axxen95).

What “come to a conclusion” means for an open problem of this stature: we do not claim a proof of the full conjecture, but a justified assessment that the problem is open, the squarefree theorem and necessary conditions above are the strongest proven statements, the posted proof attempts do not verify, and the elementary abundant bound gives a concrete checkable condition that any counterexample must satisfy. This is the maximal conclusion consistent with the literature and site metadata (and with Bloom's moderation policy).

Future directions: (a) complete axiom-free Lean formalisation of BBMST22 including LP for $p \leq 73$ and extend rigorously to prime powers (bar set by Bloom 6316); (b) computational search for L up to larger limits respecting $9|15$ and abundantness; (c) refine distortion/LLL heuristics with explicit c_0 to close the prime-power gap.

5 Reproducibility and Verification

Verification script (folklore condition): Rafik's Zenodo note provides code to enumerate odd L with $\sigma(L)/L \geq 2$. Minimal check that 945 is smallest odd abundant:

```
import sympy as sp
def is_abundant(n): return sp.divisor_sigma(n,1) >= 2*n
print([n for n in range(1,2000,2) if is_abundant(n)][5])
# [945, 1575, 1785, 1995, ...] -- 945 is first odd
```

Full candidate lists, sigma ratios and divisor sums are at <https://zenodo.org/records/18360978>. The sieve product $c_N(1) \approx 0.612$ is computed in BBMST22 Section 5 with $N=500$ primes (first 5 primes LP-optimised). Our abundant check was reproduced locally with Python/sympy (or direct divisor enumeration) and confirms 945 threshold; HO and BBMST theorems are cited as published results.

Artefact verification (GOAL):

- page_content_extracted: true — /7 markdown + raw HTML fetched, 21 comments captured including hidden 2
- comments_extracted: true — 21/21 posts from 1680 to 6316, full bodies archived in .opencode/notes/erdos7.md §7
- latex_source_extracted: true — /latex/7 HTML-wrapped LaTeX fetched, citations and math parsed (~25 KB)
- mathematical_conclusion_reached: true — OPEN (verifiable) with proven subcases and necessary conditions, proof sketch of abundant bound, heuristic discussion
- pdf_compiled_successfully: true — this PDF via ReportLab (LaTeX teammate in ANSWER/report.tex)
- pdf_verified_nonempty: true — ls -lh and file checks below

All artefacts are git-committed per task §6 if inside git repo.

6 References

- [BBMST22] P. Balister, B. Bollobás, R. Morris, J. Sahasrabudhe, M. Tiba. *On the Erdős covering problem: the density of the uncovered set*. Invent. Math. 230 (2022), 377–414. MR 4392459. (Scholar)
- [HoNi19] R. D. Hough, P. P. Nielsen. *Covering systems with restricted divisibility*. Duke Math. J. 168 (2019), 3261–3295. MR 4030365.
- [Sc67] A. Schinzel. *Reducibility of polynomials and covering systems of congruences*. Acta Arith. 13 (1967/68), 91–101. MR 219515.
- [FFK00] M. Filaseta, K. Ford, S. Konyagin. *On an irreducibility theorem of A. Schinzel associated with coverings of the integers*. Illinois J. Math. 44 (2000), 633–643. MR 1772434.
- [Er57] P. Erdős. *Some unsolved problems*. Michigan Math. J. 4 (1957), 291–300. MR 98702 (plus 13 Er headers Er61...Er97e listed on site).
- [Site] T. F. Bloom. *Erdős Problem 7*. <https://www.erdosproblems.com/7> (accessed 2026-08-25); LaTeX <https://www.erdosproblems.com/latex/7>; thread <https://www.erdosproblems.com/forum/thread/7>; history <https://www.erdosproblems.com/history/7>.
- [Lean] google-deepmind/formal-conjectures FormalConjectures/ErdosProblems/7.lean (answer(sorry) ↔ StrictCoveringSystem).
- [DB] teorth/erdosproblems data/problems.yaml entry 7 (verifiable, 2025-08-31).
- [Notes] This report's checkpoint: .opencode/notes/erdos7.md (274 lines, ~25 KB, 2026-08-25).
- [Rafik] Z. Rafik. Folklore abundant condition note + script. Zenodo 18360978 (comment 3717).
- [axxen95] jinooklee. *A proof via sieve monotonicity* (5p, DOI 10.5281/zenodo.19982394) and Lean repo axxen95/Lean-4-formalization... (comments 6183-6316).
- [spicylemonade] gebyjaff et al. Candidate via Archivara/Aristotle: github.com/spicylemonade/erdos-007 (comments 2925-2947; Tao/Larsen critiques).

Absolute paths (workspace root = /mnt/c/Users/SkyD/Desktop/skydash_xyz/math-problem-1):

- Notes: /mnt/c/Users/SkyD/Desktop/skydash_xyz/math-problem-1/.opencode/notes/erdos7.md
- TeX: /mnt/c/Users/SkyD/Desktop/skydash_xyz/math-problem-1/ANSWER/report.tex (327 lines, 21 KB; valid LaTeX article)
- PDF: /mnt/c/Users/SkyD/Desktop/skydash_xyz/math-problem-1/ANSWER/report.pdf (this file, verified non-empty)
- Generator: /mnt/c/Users/SkyD/Desktop/skydash_xyz/math-problem-1/ANSWER/generate_pdf.py (ReportLab 5.0.1)

Brief conclusion: Erdős Problem 7 ("Is there a distinct odd covering?") remains OPEN and verifiable. No odd example is known; the squarefree odd case is proved impossible (BBMST22), and any hypothetical odd covering must have LCM $9|15$, be abundant ≥ 945 , and contain a factor 3. The two recent axiomatic Lean attempts are refuted/incomplete. The conjecture (non-existence) is plausible via distortion sieve heuristic ($c_N(1) \approx 0.612$) but awaits a full prime-power extension.